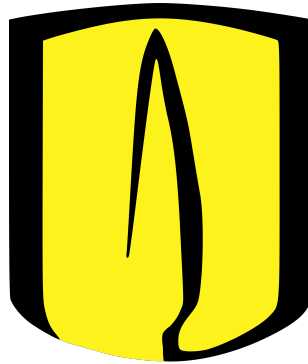


Universidad de los Andes  
Department of Computer and Systems Engineering



Graduation Project Proposal  
Bachelor's in Computing and Systems Engineering

Tomas Sierra Sanchez

Advisor: Valérie Gauthier Umaña  
Co-Advisor: Juan Fernando Pérez

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# 1 State of the art

When doing research or working in a field that involves other people's data it is crucial to set clear boundaries as to how that data is going to be collected, managed and distributed. Nowadays our data is one of our most important asset thus it becomes a necessity to create different tools that allow us to set such boundaries, one of such tools is differential privacy.

## 1.1 Differential Privacy

Differential privacy describes a promise that is made by a data holder or curator to a data subject: "You will not be affected, adversely or otherwise, by allowing your data to be used in any study or analysis, no matter what other studies, data sets, or information sources, are available".[1] In other words differential privacy is the process that either a data curator or a data holder must do in order to ensure that a single individual's data cannot be extracted from the study as a whole.

**Definition 1.1.**  *$(\epsilon, \delta)$ -Differential privacy [1]. A randomized algorithm  $\mathcal{M}$  will provide  $(\epsilon, \delta)$ -Differential privacy (henceforth  $(\epsilon, \delta)$ -DP) if for any two outputs on two different databases  $D$  and  $D'$  differing by at most one record, and for all possible outputs  $A$ :*

$$Pr[\mathcal{M}(D) \in A] \leq e^\epsilon Pr[\mathcal{M}(D') \in A] + \delta$$

In this study we are going to mostly focus around the concept of **Central differential privacy** which states that it is the responsibility of the curator (i.e a central party that receives all the raw data from the participants and adds privacy to them) to assure that the data is going to be differentially private.

## 1.2 Quantum Information and Computation

Quantum information and computation is rapidly becoming mainstream as a response to the growing necessity of larger and faster computation. Quantum computing significantly advances computational capabilities, performing exponentially faster computation than classical computers.[2]

To preface the state of quantum information these are some important definitions that will be used further in this proposal:

**Definition 1.2.** *Quantum states: A quantum state of the system  $X$  is described by a density matrix  $\rho$  whose entries are complex numbers and whose indices (for both its rows and columns) have been place in correspondence with the classical state set  $\Sigma$*

**Definition 1.3.** *Mixed state: A mixed state of a quantum system is an ensemble*

$$\mathbb{E} = \{(p_1, |\psi_1\rangle), \dots, (p_k, |\psi_k\rangle)\}$$

*which in turn can be represented by the density operator*

$$\rho_{\mathbb{E}} = \sum_i p_i |\psi_i\rangle \langle \psi_i|$$

*thus the resulting  $\rho_{\mathbb{E}}$  will be the density matrix representation of the ensemble.*

**Definition 1.4.** *Quantum Channel [3]: A quantum channel describes a linear mapping over density matrices.*

Quantum channels are important as they are the generalization of an operation over a quantum state and in the case of this project we need to use operations such as the depolarizing channel which need this general definition of a channel to be understood.

**Definition 1.5.** *Trace distance: It is the distance between two density operators and it is defined by:*

$$\tau(\rho, \sigma) = \frac{1}{2} \text{Tr}|\rho - \sigma|$$

where  $\text{Tr}(x)$  is the trace of the density matrix  $x$

**Definition 1.6.** *Quantum measurement: It is the way to extract information from a quantum system. Mathematically, a measurement is modelled as a set of matrices  $M = \{K_m\}$  with the condition of*

$$\sum_m K_m^\dagger K_m = I$$

with  $I$  being the identity of the density operator (i.e for a qbit  $I = 2 \times 2$  identity matrix)

When talking about quantum differential privacy below it won't really matter what the quantum state collapses into. Given this liberty we can define a POVM (Positive Operator-Valued Measure) which is a simplification over the measurement where we get that  $M = \{M_m\}$  with  $M_m = K_m^\dagger K_m$   $\forall m$  since  $p_m = \text{Tr}(K_m \rho K_m^\dagger) = \text{Tr}(M_m \rho)$

**Definition 1.7.** *Quantum noise: It is a inherent physical process that occurs on the quantum state that causes it to change in some way.*

For the purposes of this investigation we are interested in three types of noise: General amplitude dampening, Phase-angle dampening and Depolarizing noise. Each one of these types represent a unique physical process which is summarized in greater detail in [2] and [4]. However for this proposal it is only necessary to note that these processes will randomize the quantum state by a certain degree.

### 1.3 Quantum Differential Privacy

Given the inherent noise of the current quantum framework, one could argue that it is a perfect scenario for differential privacy. In fact, the promise that the classical DP curator makes to the data holder does will not change when in the quantum setting. However, the means that the curator will take to reach that promise do.

With that in mind it was important to figure out what the current investigative and academic situation was with regards to a possible quantum differential privacy application was.

The results of this investigation are compiled in Figure 1.

As the reader can observe the investigation yielded positive results as there were many academic papers that not only mentioned the topic but gave valuable research and insight on possible angles the topic could leap into. Among the most interesting ones are: [8] in which a hybrid quantum machine learning framework was proposed to allow differential privacy in the model. [2] which tested different quantum differential privacy mechanisms along with proposing its own **Lifted** quantum differential privacy, which enhanced the randomness induced in the quantum states. And finally in the study of Zhou and Ying [4] they define a framework of differential privacy in the quantum setting. Moreover they examine how different noise mechanisms change the protection of that privacy budget  $\epsilon$ . Thus giving the following definition:

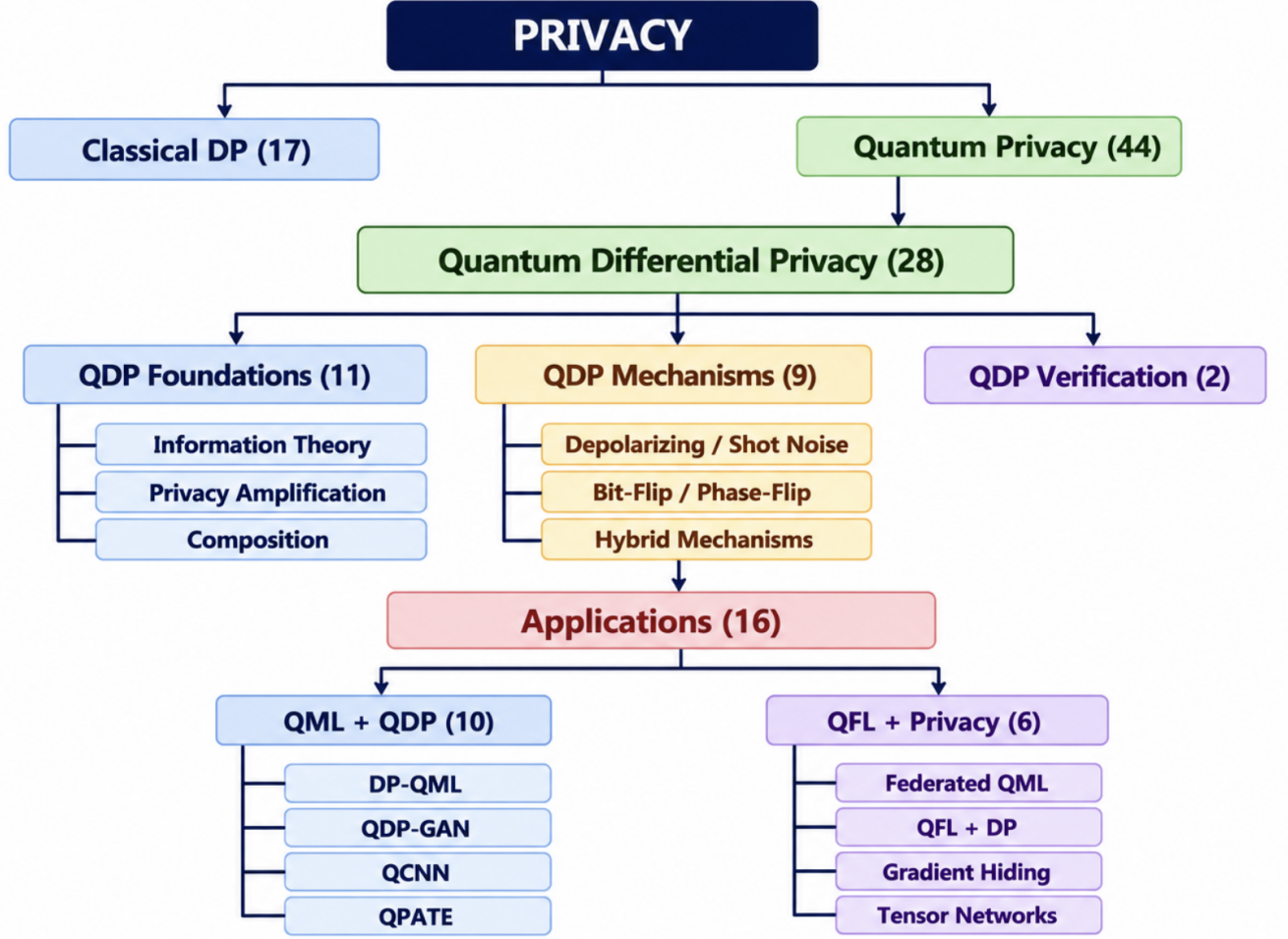


Figure 1: State of the art of quantum differential privacy.

**Definition 1.8.** (Quantum differential privacy [4]). Let  $d \in (0, 1]$  and  $\epsilon, \delta > 0$  be three constants. A quantum operation  $\mathcal{E}$  is  $(\epsilon, \delta)$ -differentially private if for every POVM  $M = M_m$ , for all outputs  $S \subseteq \text{Out}(M)$  and for all inputs  $\rho, \sigma$  such that  $\tau(\rho, \sigma) \leq d$ , it holds that:

$$\Pr[\mathcal{E}(\rho) \in_M S] \leq e^\epsilon \Pr[\mathcal{E}(\sigma) \in_M S] + \delta$$

There are many ways to obtain that epsilon in theory however Zhou and Ying proposed the following algorithm to obtain a quantity called proportional distance which can help obtain  $\epsilon$ .

**Definition 1.9.** (Proportional Distance [4]). For two density operators  $\rho$  and  $\sigma$ , their proportional distance is defined as:

$$PD(\rho, \sigma) = \sup_{\{M_m\}, m} \max \left( \ln \frac{q_m}{p_m}, \ln \frac{p_m}{q_m} \right)$$

where the supremum is over all POVMs  $\{M_m\}$  and all possible outcomes  $m$ ,

$$p_m \equiv \text{Tr}[\rho M_m], \quad q_m \equiv \text{Tr}[\sigma M_m]$$

are the probabilities of obtaining outcome  $m$  when the measurement is performed on  $\rho$ ,  $\sigma$ , respectively, and we make the conventions  $\frac{0}{0} = 1$ ,  $\frac{1}{0} = +\infty$ ,  $\ln 0 = -\infty$ ,  $\ln +\infty = +\infty$ .

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**Algorithm 1:** Computing  $dPD$

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1 Function  $DPD(\rho, \sigma, \theta)$ 
  //  $\rho$  and  $\sigma$  are two density operators in Hilbert space  $\mathcal{H}$  and  $\theta$  is the desired
  accuracy.
2 Let  $n$  be the dimension of the Hilbert space  $\mathcal{H}$ .
3 Set real number  $max = 0, \lambda = \infty$ .
4 while  $|max - \lambda| > \theta$  do
5    $\lambda = max$ 
6    $max = \max_i(\rho_{ii}/\sigma_{ii})$ 
7    $\eta = \rho - max * \sigma$ 
8   Diagonalize  $\eta$ :  $\eta = PDP^\dagger$ 
   //  $P$  is unitary and  $D$  is diagonal matrix.
9    $\rho = P^\dagger \rho P$ 
10   $\sigma = P^\dagger \sigma P$ 
11 end
12 return  $\ln max$ 

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One thing that was apparent during the investigation of the state of the art was that the quantum circuits or simulators used were ambiguous to say the least and I felt there was some possibility to improve on the clarity of the tests towards the user. Secondly it was clear that the theory and experiments by themselves were very solid. But again, I felt there was space for improvements. These improvements became the cornerstones of this project.

## 2 Objectives

### 2.1 General Objective

The development of this project will tackle two main objectives: The first will involve an adequate understanding of quantum information and computation as well as the theoretical framework of QDP. The second will be to empirically validate the theoretical  $(\epsilon, \delta)$ -DP guarantees provided by quantum noise mechanisms such as the depolarizing channel, phase-amplitude dampening and GAD, by comparing the privacy budget derived in [4] with the empirical privacy budget measured on a practical quantum simulation on the PennyLane framework.

### 2.2 Specific Objectives

- Gain an understanding of the classical definition of differential privacy, as well as quantum information and computation to understand quantum differential privacy.
- Create an experiment on the quantum simulation framework PennyLane which will test  $(\epsilon, \delta)$ -DP over a real dataset [6]. These results will be explored in further detail in the project's paper.
- Establish whether the theoretical budgets proposed in [4] and [2] agrees or disagrees with the results of the experiments in the simulations for each different noise mechanism.

### 3 Description of the Experiment

The experiment will follow the following stages:

1. Encode the dataset  $D$  with the following rules.
  - (a) For categorical variables a set angle will encode each of the three possible categories.
  - (b) For non categorical variables each one will be encoded by dense angle encoding thus giving two variables per qbit.
2. Encode the dataset  $D'$  which is the same dataset  $D$  with one participant removed with the same rules as **1**
3. Simulate a process over the qbits that result from **1** and **2** such as aggregation. The result from this process on each qbit are going to be the density matrices called  $\rho$  and  $\sigma$  respectively.
4. Insert noise into  $\rho$  and  $\sigma$
5. Run **Algorithm 2.**[4] on  $\rho$  and  $\sigma$  to get the experimental  $\epsilon$  for that specific run
6. Run the experiment for all  $D'$  to get the experimental  $\epsilon$  for the whole dataset.

### 4 Expected Results

To achieve this the specific results are as follows:

1. Prove that in practice the privacy budget will be preserved within reasonable proximity to the theoretical budget in an experimental setup for the different noise injections sources and classical data sources.
2. Provide the source code for the experimental setup in PennyLane with python.
3. Write a scientific article in IEEE format. The information within this article should demonstrate my understanding of the topic as it will contain the whole process of investigation, creation and result analysis of this project.
4. Design a poster which will summarize the most important parts of the project.

### 5 Ethical Considerations

This investigation will use an adaptation [7] of the Statlog German Credit Data dataset [6], originally compiled by Prof. Hans Hofmann in 1994 which has a public CC BY 4.0 license thus allowing the use of the dataset for any purpose as long as the appropriate credit is provided.

This dataset was selected as it encompasses the exact necessity that differential privacy is trying to protect: Given sensitive data from individuals (sex/marital status, foreign worker status, etc.), what is the privacy budget that can be offered to the individuals in exchange of the use of their information in the dataset.

## 6 Activity Calendar

| Week #                                                                            | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|-----------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|
| Activity                                                                          |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
| Investigation on the viability of the project and scope                           |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
| Write the proposal of the project                                                 |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
| Search and data analysis of the experimental data to be used in the experiments   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
| Write the introductory section of the paper (Introduction, Theoretical Framework) |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
| Real data experiment development in PennyLane                                     |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
| Analysis of the first two experiment's results                                    |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
| Write the first revision of the paper                                             |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
| Adjustments to the experiments with respect to the result of the analysis         |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
| Write the second revision of the paper                                            |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
| Write and design the other deliverables (Poster, Project document, etc)           |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
|                                                                                   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |

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